

Predicting Customer Booking Completion

Using customer booking data to rank booking probability
and support targeted marketing decisions

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Feature engineering * Predictive modelling * Model comparison

Business Problem

Challenge



Thousands of customers start booking flights every day

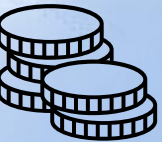
Many leave before completing their reservation

Objective



Predict the probability of booking completion - to identify customers who may still be influenced by marketing

Business Value



- ☐ Higher booking completion
- ☐ More efficient marketing spend
- ☐ Improved marketing ROI

Origin and Destination Airports

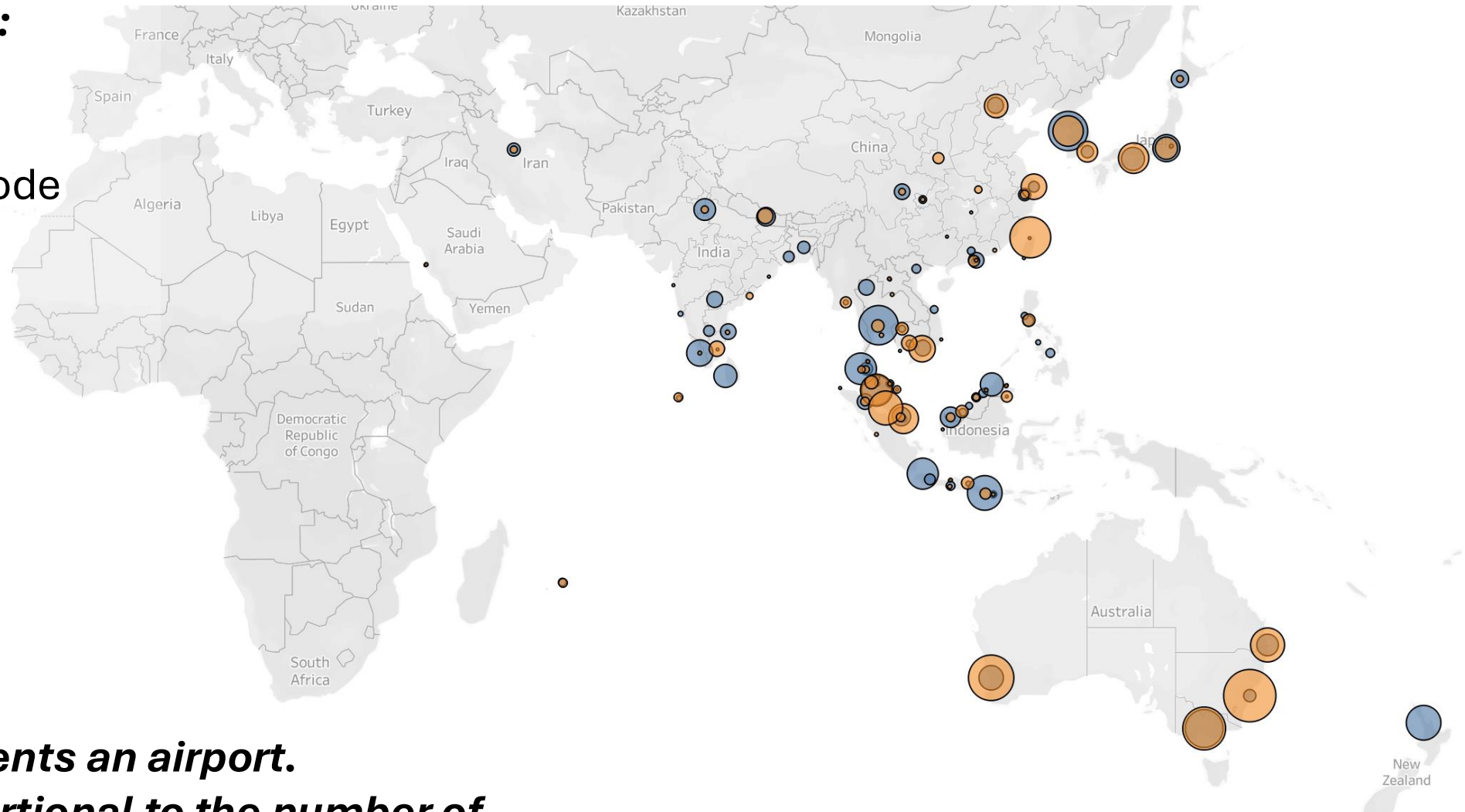
Feature Engineering:

Original variable:

Route (IATA airport code format)

Derived variables:

Origin Airport and
Destination Airport



Each bubble represents an airport.

Bubble size is proportional to the number of bookings originating from or arriving at that airport.

Number of Bookings by Booking-Origin Country



Colour intensity represents the number of booking records originating from each country.

Country Grouping for Predictive Modelling

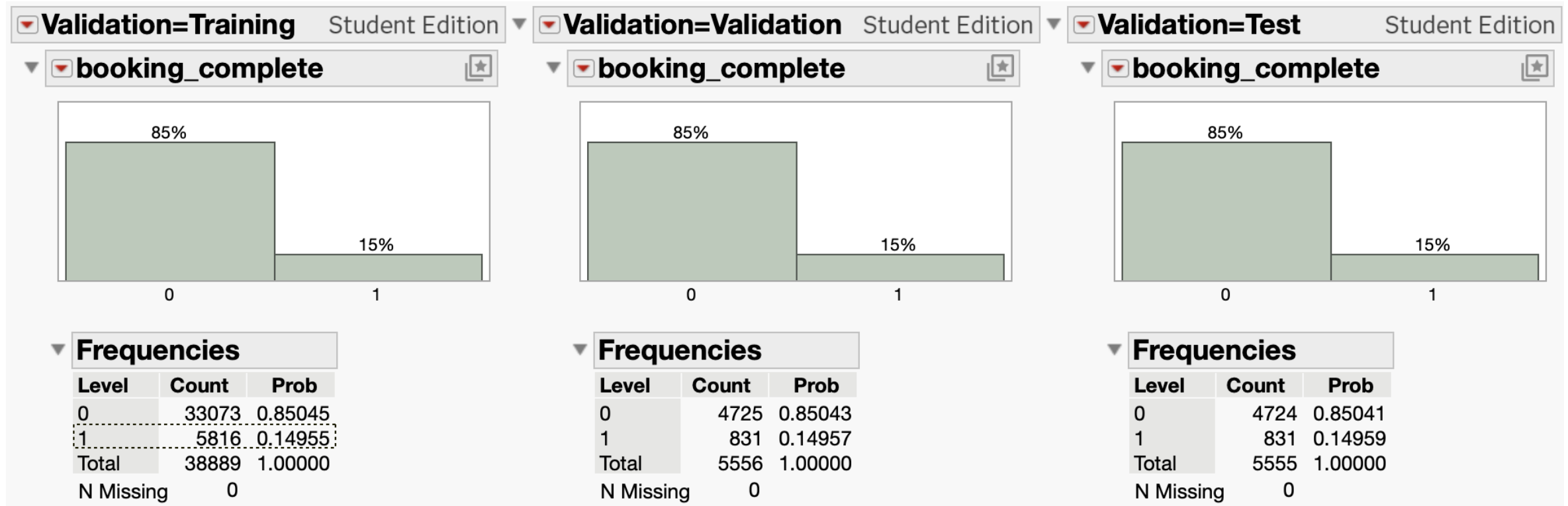
Country Grouping	
Australia	17,872
Malaysia	7,174
South Korea	4,559
Other Country	4,219
Japan	3,885
China	3,387
Indonesia	2,369
Taiwan	2,077
Thailand	2,030
India	1,270
New Zealand	1,074
Unknown Country	84

Feature Engineering Decision:

- ❑ Most bookings originate from Australia, followed by Malaysia, South Korea and Japan.
- ❑ Booking activity is concentrated in the Asia–Pacific region, indicating that most travel demand in this dataset comes from this market.
- ❑ Countries with fewer than 1,000 booking records were combined into the "Other Country" category to reduce model complexity while preserving the major markets.

Country grouping was used as a feature engineering step before building the predictive models.

Stratified Data Partitioning and Class Distribution



The target variable is imbalanced ($\approx 15\%$ completed bookings). Therefore, a stratified 70/15/15 partition was used to preserve the class distribution across the Training, Validation and Test sets.

Class Weighting to Address Class Imbalance

Inverse-frequency class weights:

$$w_i = N / (K \times N_i)$$

where:

- N = total number of observations
- K = number of classes
- N_i = number of observations in class i

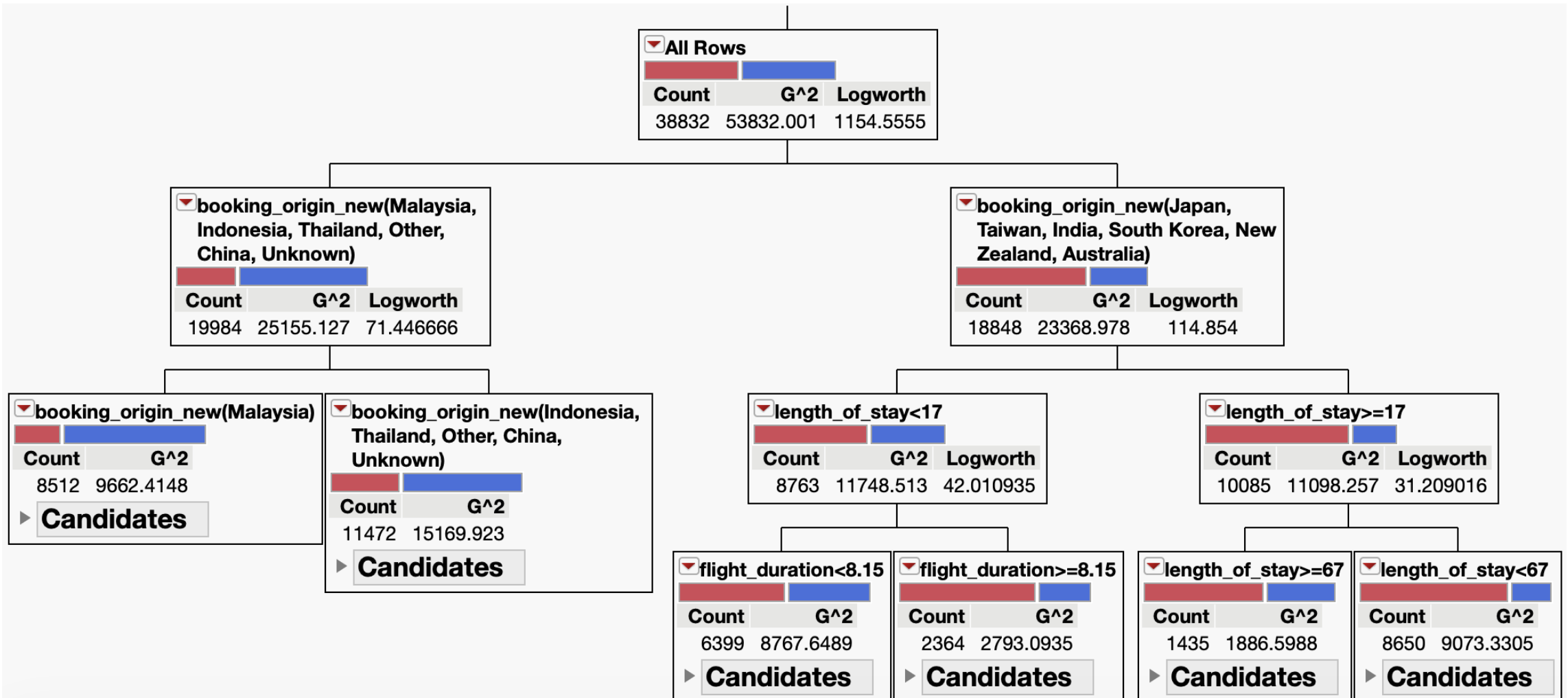
For this dataset:

$$w_{\text{completed}} = 3.333$$

$$w_{\text{not completed}} = 0.588$$

Higher weights were assigned to the minority class so that completed bookings had a greater influence during model training.

Customer Segmentation Using Decision Tree

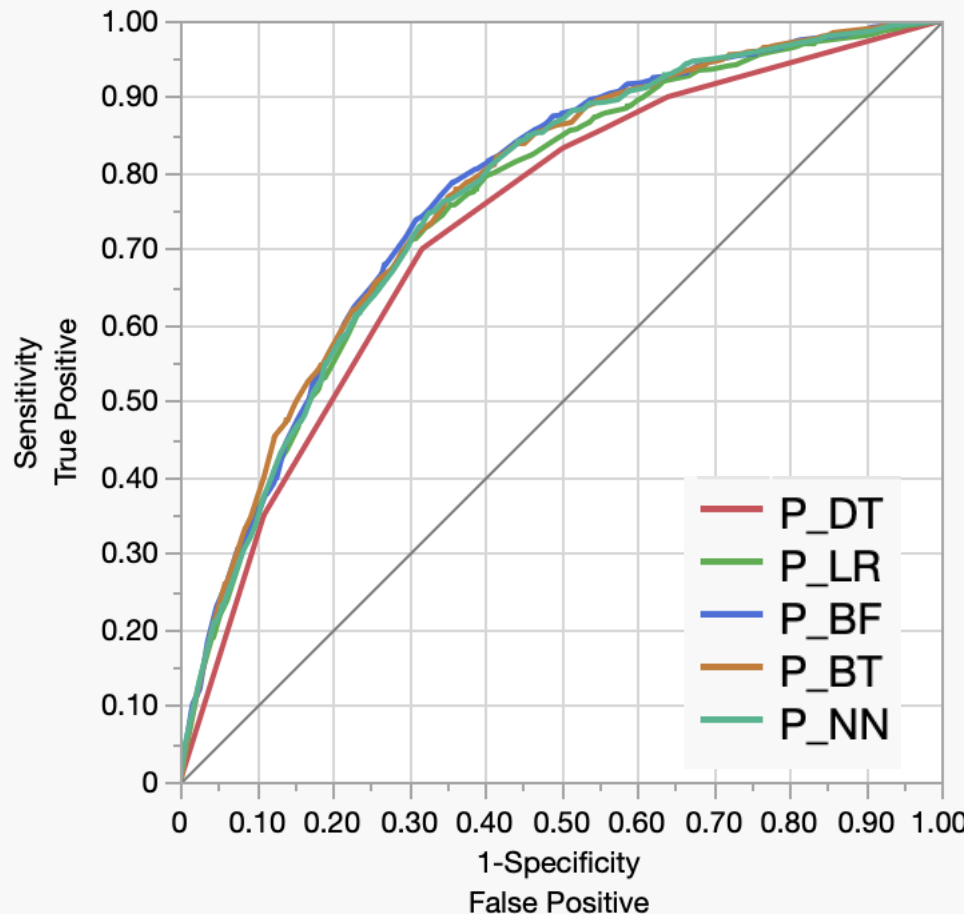


Booking-origin country is the primary factor associated with booking completion, followed by travel characteristics such as length of stay and flight duration.

Predictive Model Comparison

Model performance of the Test partition:

ROC Curve for booking_complete=1

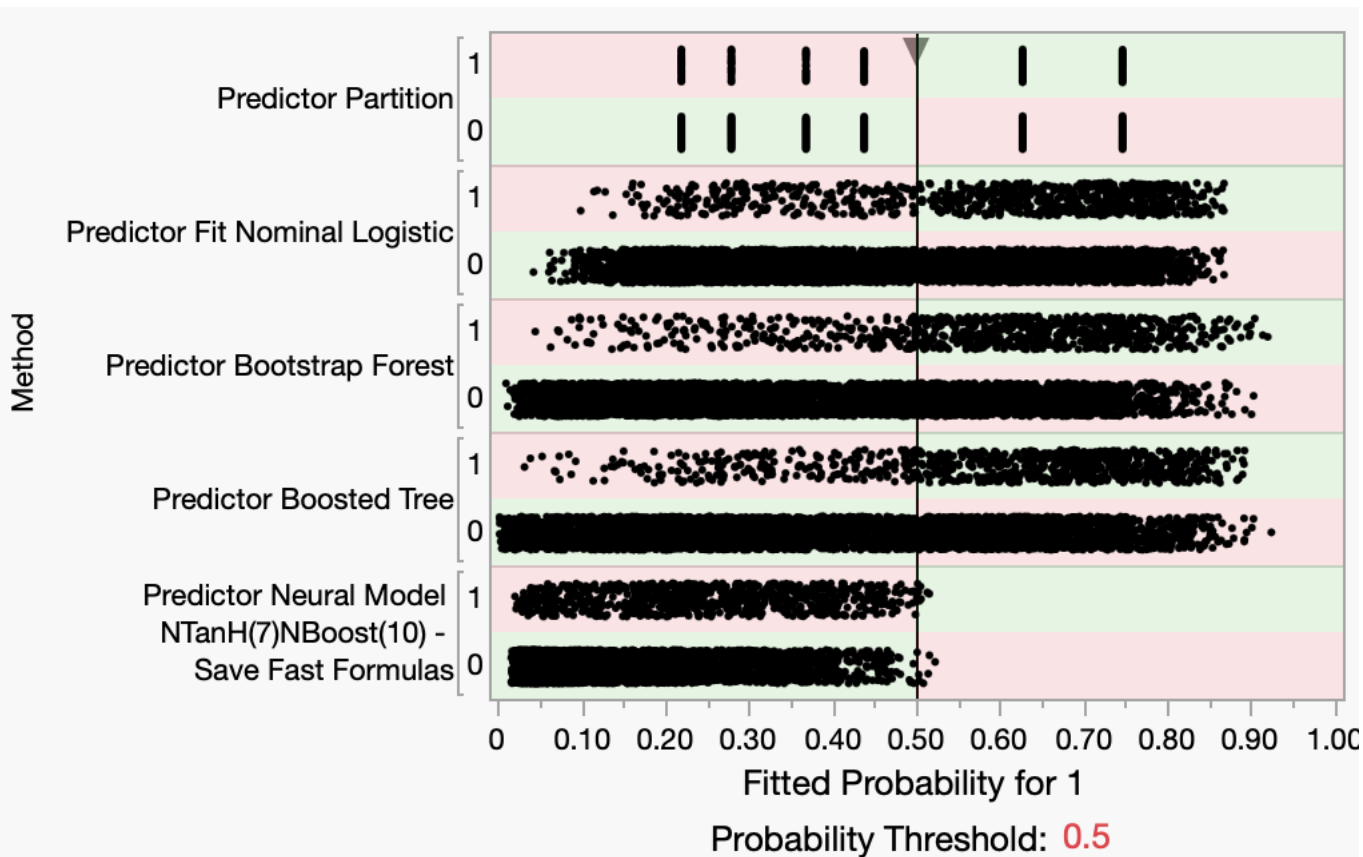


Model	ROC AUC
Decision Tree	0.731
Logistic Regression	0.757
Bootstrap Forest	0.771
Boosted Tree	0.770
Neural Network	0.765

Several predictive models were evaluated using the independent test dataset. Bootstrap Forest achieved the highest ROC AUC (0.771) and was selected as the best-performing model for further threshold analysis.

From Prediction to Decision

Using Predicted Probabilities for Decision-Making:



- ☐ The model estimates the **probability** that a customer will complete the booking.
- ☐ The **probability threshold** determines which customers are selected for marketing intervention.
- ☐ The **optimal threshold** depends on **campaign objectives**, available budget, and the expected value of additional bookings.

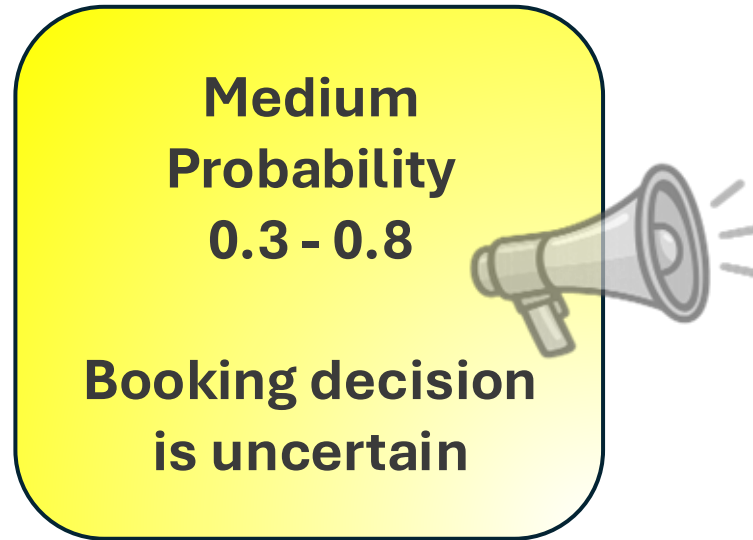
The threshold is a business decision, not a model parameter.

Business Recommendations

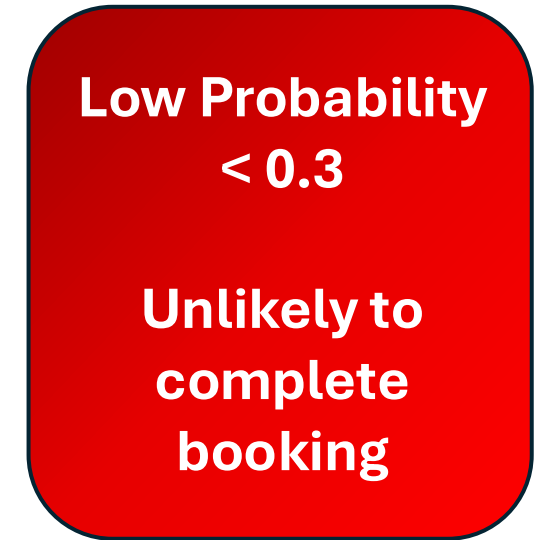
Targeting Customers with Medium Booking Probability:



No additional
marketing required



**Targeted marketing
campaign**



Limited or low-cost
marketing

The probability threshold should be adjusted according to campaign cost and expected return on investment (ROI).

Final Recommendation

- ☐ **Predict booking probability.**
- ☐ **Focus marketing on customers with medium booking probability.**
- ☐ **Adjust the probability threshold according to campaign cost and expected ROI.**